

AP Calculus BC CA #3

Due date: May 18, 2026

Mathematics & Art Through the Ages Outline Submission

Your Name: Aven Mitchell Partner Name: (if applicable)

1. The art piece

a. Who is the artist? Daniel Chester French.

b. What was the inspiration for the piece of art? The direct inspiration was Spencer Trask and Saratoga Springs itself. The sculpture was commissioned as part of a memorial to Trask, a businessman and civic leader who helped restore Saratoga Springs as a health resort. French represented that inspiration through a winged female figure called The Spirit of Life, holding a bowl of flowing water to connect the memorial to Saratoga's famous mineral springs and holding a pine branch associated with Trask. In other words, the piece was inspired by Trask's life of public service, the healing waters of Saratoga, and the symbolic idea of life, renewal, and spiritual energy.

c. What makes the piece of art what it is? It is a larger-than-life bronze fountain sculpture. The figure is winged, draped, and shown with raised arms, holding a pine branch and a bowl of water. The water, reflecting pool, stone setting, and bronze figure make the work both sculpture and public architecture. Its meaning depends on allegory: the woman is not meant to be one ordinary person, but a symbol of life, inspiration, and renewal.

d. What is important about the piece of art? It is important because it shows how early twentieth-century American public art used classical forms, proportion, and symbolic figures to honor civic leaders. French was one of the most important American sculptors of the period, and this work shows the same Beaux-Arts monument style that shaped many public memorials.

e. Why did you choose the piece of art? I chose the piece because Ramanujan believed that a goddess helped him find solutions to mathematical problems in his dreams. The sculpture represents a divine or spiritual female figure, so it works as a visual connection to the idea of Namagiri giving inspiration to Ramanujan.

f. Any other interesting information you find about the piece of art. The sculpture was cast in November 1914 and dedicated on June 26, 1915. Henry Bacon, the architect later known for the Lincoln Memorial, worked on the setting. A related preliminary model called Spirit of the Waters is in the Metropolitan Museum of Art collection, where it is described as an early version of the final Spirit of Life design.

2. What mathematics was used in art for the time period?

a. Mathematics in art for the time period During the early 1900s, artists used mathematics in two different ways. Traditional public sculpture still depended on proportion, symmetry, perspective, scale, volume, and careful measurement. At the same time, modern movements such as Cubism and geometric abstraction made geometry more visible by breaking objects into planes, angles, cylinders, cones, and cubes. In education and industry, many people also focused less on abstract reasoning alone and more on practical mathematical skills that could be used in engineering, manufacturing, design, and everyday work.

b. Give specific examples The Spirit of Life uses balance, bilateral symmetry, scale, and geometric planning in the fountain, pool, pedestal, and figure. Cubist artists such as Pablo Picasso and Georges Braque used geometric solids and multiple viewpoints instead of one fixed Renaissance perspective. Architects and sculptors used measurement, ratios, and structural geometry to make public monuments stable, readable from a distance, and harmonious with their settings.

3. What influenced the art and mathematics of the time period?

a. Politics - Art Progressive Era civic reform and the City Beautiful movement encouraged cities to build parks, monuments, fountains, and public memorials. Wealthy patrons and local governments supported works that made cities look orderly, moral, and cultured. World War I also affected the period by making public memory and national identity more important themes in art.

a. Politics - Mathematics Governments, businesses, and schools needed people who could calculate accurately for engineering, manufacturing, finance, navigation, statistics, and military work. This pushed mathematics education toward practical arithmetic and applied problem solving, even while university mathematicians continued developing more abstract fields such as number theory.

b. Historical facts or historical influences - Art Industrialization changed cities and created demand for public spaces such as parks and memorials. Artists trained in Beaux-Arts methods still looked to classical Greece and Rome for idealized human figures, but modern artists were also reacting against old rules of perspective and realism. New materials and foundries made large bronze public sculptures possible.

b. Historical facts or historical influences - Mathematics Ramanujan worked during a period when mathematics was becoming more international. He developed many ideas independently in India, then began corresponding with G. H. Hardy in 1913 and worked at Cambridge beginning in 1914. In schools, the early twentieth century also saw the rise of educational psychology, including Edward Lee Thorndike, and later debates over reform movements such as New Math.

c. Mathematical needs of the time The growth of industry meant workers needed more applicable math skills instead of only abstract skills. There was more emphasis on basic skills like arithmetic, measurement, estimation, formulas, and real-world applications. At the same time, advanced mathematics was needed for physics, engineering, statistics, finance, and the new technologies of the twentieth century.

4. How has the mathematical topic evolved over time?

a. Explain the chosen topic The chosen topic is number theory, especially Ramanujan's work with integer partitions. Number theory studies the properties of whole numbers. A partition of a number is a way to write it as a sum of positive integers. For example, 4 has five partitions: 4, 3 + 1, 2 + 2, 2 + 1 + 1, and 1 + 1 + 1 + 1.

b. Who was given credit for the introduction of the topic? Number theory has ancient roots, including Greek, Indian, and other early mathematical traditions, so no one person introduced the whole field. Euler is often credited with major early work on partitions. Ramanujan is credited with major breakthroughs in partition congruences and, with Hardy, an important asymptotic formula for the partition function.

c. Is the topic still relevant today? Yes. Number theory is still one of the most active areas of mathematics. It is used in cryptography, computer science, algorithms, coding theory, combinatorics, and mathematical physics. Ramanujan's ideas also continue to influence modern research in modular forms, mock theta functions, and partitions.

d. Describe the evolution of the topic Number theory began with basic questions about counting numbers, divisibility, primes, and factorization. Mathematicians such as Fermat, Euler, and Gauss made it more systematic. In the late nineteenth and early twentieth centuries, analytic number theory connected whole-number problems with calculus and complex analysis. Ramanujan pushed this forward with discoveries about partitions, infinite series, continued fractions, and modular equations. After his death, mathematicians continued proving and expanding results from his notebooks, including work related to modular forms and mock theta functions.

e. In what field is the math topic used? It is used in pure mathematics, computer science, cybersecurity, cryptography, coding theory, combinatorics, physics, and any field that studies patterns in discrete structures.

Sources - MLA format

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